# Pablo M. Berná

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## Summary

PhD in Mathematics with International Mention (UAM), specialized in Functional Analysis and Approximation Theory. My research career focuses on the study of greedy algorithms in Banach and quasi-Banach spaces, with more than 30 publications in high-impact journals. Currently, I combine my position as Tenure Track Professor at CUNEF University with active institutional management responsibilities, holding leadership positions both at the Polytechnic School of CUNEF and at the Spanish Royal Mathematical Society (RSME). My commitment to academic excellence is reflected in the supervision of doctoral theses and continuous participation in competitive national research projects.

## Education

– **PhD in Mathematics** (Cum Laude, International Mention)
Universidad Autónoma de Madrid, 2017–2020.
Title: *"Thresholding greedy algorithms in Banach spaces"*.

– **Master's Degree in Education: Mathematics and Computer Science**
International University of Valencia, 2018–2019.

– **Master's Degree in Mathematical Research**
University of Valencia, 2014–2015.

– **Bachelor's Degree in Mathematics**
University of Alicante, 2009–2014.

## Management, Academic Service and Accreditations

– **Deputy Director of the Polytechnic School**, CUNEF University (2024–Present).

– **Deputy Secretary of the Spanish Royal Mathematical Society** (2025–Present).

– **Member of the Editorial Board** of *Demonstratio Mathematica* (Q1, JCR) and *Journal of Mathematical Analysis.*

– **ANECA Accreditations:** Associate Professor (Profesor Titular) and Tenured Contracted Professor (Profesor Contratado Doctor).

– **Research Evaluation Period (Sexenio):** Recognized six-year research period 2018–2023.

## Scientific Publications

[1] F. Albiac, J. L. Ansorena, M. Berasategui and P. M. Berná. Conditional bases with Property (A). Accepted in Constructive Approximation (March 2026). Available at: [PDF].

[2] P. M. Berná. Step-Size Decay and Structural Stagnation in Greedy Sparse Learning. Mathematics 2026, 14(6), 967.

[3] P. M. Berná, A. García, *The Weak TGA in p-Banach spaces.* Dolomites Research Notes on Approximation, 19(1), 48-55 (2026).

[4] M. Berasategui, P. M. Berná, H. V. Chu, *On sequential greedy-like bases.* Ann. Funct. Anal. **16**, 36 (2025).

[5] P. M. Berná, H. V. Chu, E. Hernández, *On approximation spaces and Greedy-type bases.* Ann. Funct. Anal. (2025) 16:7

[6] M. Berasategui, P. M. Berná, D. González, *Approximation by polynomials with constant coefficients and the Thresholding Greedy Algorithm.* Carpathian J. Math. **41**, no. 1, (2025).

[7] P. M. Berná, A. Falcó, *Universality for non-linear convex variational problems*, J. Nonlinear Var. Anal. **9** (2025), 161-177.

[8] M. Berasategui, P. M. Berná, H. V. Chu, *On consecutive greedy and other greedy-like type of bases.* Res. Math. Sci **11**, 61 (2024).

[9] P. M. Berná, A. García, D. González, *A note on 1-semi-greedy bases in p-Banach spaces with* $0 < p \leq 1$. Demonstratio Mathematica, vol. 57, no. 1, 2024, pp. 20240047.

[10] M. Berasategui, P. M. Berná, *Greedy-like bases for sequences with gaps.* Banach J. Math. Anal. **18**, 14 (2024).

[11] P. M. Berná, D. González, *Non-linear approximation by* 1*-greedy bases.* J. Math. Anal. App. **531**, no 1. (Part 1), (2024), 127769.

[12] M. Berasategui, P. M. Berná, H. V. Chu, *Extensions and new characterizations of some greedy-type bases*, Bull. Malay. Math. Sci. Soc. **46**, 84 (2023).

[13] F. Albiac, J. L. Ansorena, M. Berasategui, P. M. Berná, S. Lasalle, *Bidemocratic bases and their connections with other greedy-type bases.* Constr. Approx. **57** (2023), 125-160.

[14] F. Albiac, J. L. Ansorena P. M. Berná, *New parameters and Lebesgue-type estimates in greedy approximation*, Forum of Mathematics Sigma **10** (2022), e113, 1–39.

[15] M. Berasategui, P. M. Berná, *Extensions of democracy-like properties for sequences with gaps*, Math. Inequal. Appl. **25**, no. 4 (2022) 1155–1189.

[16] P. M. Berná, , H. V. Chu, *On some characterization of greedy-type bases*, Expo. Math. **40** (2022), 1135-1158.

[17] F. Albiac, J. L. Ansorena, M. Berasategui, P. M. Berná, S. Lasalle, *Weaker forms of unconditionality of bases in greedy approximation*, Studia Math. **257** (1) (2022), 1-17.

[18] P. M. Berná, D. Mondéjar, *A functional characterization of almost-greedy bases.* Mathematics (2021) 9(15):1827. https://doi.org/10.3390/math9151827

[19] M. Berasategui, P. M. Berná, *Quasi-greedy bases for sequences wigh gaps.* Nonlinear Analysis, **208** (2021), 112294.

[20] F. Albiac, J. L. Ansorena, P. M. Berná, P. Wojtaszczyk, *Greedy approximation for biorthogonal systems in quasi-Banach spaces.* Dissert. Math. **560** (2021), 1-88.

[21] P. M. Berná, *A note on partially-greedy bases in quasi-Banach spaces.* Studia Math. **259** (2021), 225-239.

[22] M. Berasategui, P. M. Berná, S. Lasalle, *Strong-partially-greedy bases and Lebesgue-type inequalities.* Constr. Approx. **54** (2021), 507-528.

[23] P. M. Berná, J. D. Rossi, *Nonlocal diffusion equations with dynamical boundary conditions.* Nonlinear Analysis **195** (2020), 111751.

[24] F. Albiac, J. L. Ansorena, P. M. Berná, *Asymptotic greediness of the Haar System in the $L_p$ spaces for $1 < p < \infty$*. Constr. Approx. **51**, 427—440 (2020).

[25] P. M. Berná, *Characterization of weight-semi-greedy bases.* J. Fourier. Anal. Appl. **26**, 21 (2020).

[26] P. M. Berná, Ó. Blasco, G. Garrigós, E. Hernández, T. Oikhberg, *Lebesgue inequalities for Chebyshev Thresholding Greedy Algorithms.* Rev. Matem. Complut. **33**, 695-722 (2020).

[27] P. M. Berná, D. Kutzarova, S. J. Dilworth, T. Oikhberg, B. Wallis, *The weighted Property (A) and the greedy algorithm.* J. Approx. Theory **248** (2019), 105300.

[28] P. M. Berná, A. Pérez, *A remark on approximation with polynomials and greedy bases.* J. Math. Anal. App. **478** (2) (2019), 466–475.

[29] P. M. Berná, *Equivalence between almost-greedy and semi-greedy bases.* J. Math. Anal. App. **417** (2019) 218–225.

[30] P. M. Berná, Ó. Blasco, G. Garrigós, E. Hernández, T. Oikhberg, *Embeddings and Lebesgue inequalities for greedy algorithms.* Constr. Approx. **48** (3) (2018), 415–451.

[31] P. M. Berná, Ó. Blasco, G. Garrigós, *Lebesgue inequalities for the greedy algorithm in general bases.* Rev. Mat. Complut. **30** (2017), 369–392.

[32] P. M. Berná, Ó. Blasco, *The best m-term approximation with respect to polynomials with constant coefficients.* Anal. Math. **43** (2) (2017), 119–132.

[33] P. M. Berná, Ó. Blasco, *Characterization of greedy bases in Banach spaces.* J. Approx. Theory **215** (2017) 28–39.

## Submitted Papers

[1] F. Albiac, J. L. Ansorena, M. Berasategui, P. M. Berná. Isometric renormings for greedy bases in Banach spaces, with applications to the Haar System in $L_p[0,1]$, $1 < p < \infty$. Available at: [PDF]. Submitted to Proceedings of the London Mathematical Society.

[2] P. M. Berná and A. García. Convergence Analysis of Greedy Algorithms with Adaptive Relaxation in Hilbert Spaces (submitted 2026). Available at: [PDF]. Submitted to Mathematical Modelling and Analysis.

[3] P. M. Berná, D. Freeman, T. Oikhberg and M. A. Taylor. Maximal inequalities, frames and greedy algorithms (submitted 2026). Available at: [PDF]. Submitted to Compositio Mathematica.

[4] M. Berasategui, P. M. Berná and A. García. Approximation spaces, greedy classes and Lorentz spaces (submitted 2025). Available at: [PDF]. Submitted to Constructive Approximation.

[5] M. Berasategui, P. M. Berná, H. V. Chu and A. García. Lebesgue-type estimates for greedy algorithms in quasi-Banach spaces (submitted 2025). Available at: [PDF]. Submitted to Revista Matemática Complutense.

[6] M. Berasategui, P. M. Berná, S. J. Dilworth and D. Kutzarova. Summability Methods for the Greedy Algorithm in Banach spaces (submitted 2025). Available at: [PDF]. Submitted to RACSAM.

[7] F. Albiac, J. L. Ansorena, M. Berasategui and P. M. Berná. When Greedy Approximation Breaks: Counterexamples in Quasi-Banach Spaces (submitted 2025). Available at: [PDF].

## Works in progress

[1] P. M. Berná and D. Mondéjar. Some results about 1-semi-greedy bases.

[2] P. M. Berná and D. Mondéjar. Greedy algorithms in the context of TDA.

[3] P. M. Berná and A. García. Weighted greedy bases in $p$-Banach spaces.

[4] P. M. Berná and A. García. Review of greedy algorithms in Hilbert spaces.

[5] M. Berasategui, P. M. Berná, D. Kutzarova, T. Oikhberg. Canonical basis of $\ell_p$ and $c_0$.

[6] P. M. Berná. Step-size decay in greedy sparse learning.

## PhD Supervision

- **Andrea García Pons** (Ongoing, 2024–2027)
  *Greedy Algorithms and their applications in Banach spaces.* Institution: CEU Universities. Co-direction with Miguel Berasategui.
- **David González Moro** (Ongoing, 2023–2026)
  *Greedy-type bases and nonlinear approximation theory.* Institution: CEU Universities.
- **Hung Viet Chu** (Completed, 2019–2023)
  *Relaxations of the optimality requirement on the Thresholding Greedy Algorithm for bases of Banach spaces.* Institution: University of Illinois at Urbana-Champaign (Co-advisor).

## Research Visits

– **University of Illinois at Urbana-Champaign** (USA): September–October 2017, March 2023, April 2025.

– **Isaac Newton Institute** (Cambridge, UK): July 2024.

– **University of Cambridge** (UK): February–May 2019.

– **University of Buenos Aires** (Argentina): August–October 2019.

## Participation in Research Projects

◇ **Análisis de Fourier y Aplicaciones**
Entidad: Agencia Estatal de Investigación (AEI). Reference: `PID2022-142202NB-100`.

◇ **Análisis de Fourier y Aplicaciones**
Entidad: Agencia Estatal de Investigación (AEI). Reference: `PID2019-105599GB-100`.

◇ **La interacción entre geometría y topología en espacios de Banach y sus aplicaciones**
Entidad: Ministerio de Economía y Competitividad (MINECO). Reference: `MTM2017-83262-C2-2-P`.

◇ **Interacción y aplicaciones en Análisis Funcional y Armónico**
Referencia: `20906/PI/18`, Fundación Séneca.

◇ **Análisis de Fourier, Análisis Real y Aplicaciones**
Reference: `MTM2016-76566-P`.

◇ **Interacción y aplicaciones en Análisis Funcional y Armónico**
Reference: `19368/PI/14`, Fundación Séneca.

**Exploria Teaching Innovation Project**: Innovation in the teaching of Mathematics in Economics and Business degree programs. Institution: CEU San Pablo University (2020–2021).

## Invited Talks and Conferences

I have delivered nearly 50 contributions at national and international conferences and seminars.

### Selection of Invited Talks

- *Universality for non-linear convex variational problems.*
  Congreso de Jóvenes Investigadores de la RSME 2025 - Bilbao, España (Febrero 2025).
- *Recent advances about greedy-like bases.*
  Sub-Riemannian Geometry Harmonic Analysis, PDEs and Applications, Università di Bologna, Italia (Julio 2023).
- *The bidemocracy in the world of greedy approximation.*
  Approximation and Discretization (Online), Moscú (Septiembre 2021).
- *Lebesgue-type estimates for the Thresholding Greedy Algorithm.*
  New Bridges between Mathematics and Data Science, Valladolid (Noviembre 2021).
- *La avaricia de la base de Haar.*
  EARCO 2019, Universidad de Valencia (Junio 2019).

### Some contributions (Talks, Posters and Seminars)

- *Introduction to the TGA in Banach spaces.*
  Learning Seminar in Banach spaces - University of Illinois at Urbana-Champaign, EEUU (Abril 2025).
- *Greedy-like bases for sequences with gaps.*
  Congreso Bienal de la RSME 2024 - Pamplona, España (Enero 2024).
- *Approximation spaces and greedy-like bases.*
  Foundation of Computational Mathematics, París (Junio 2023).
- *New results about Lebesgue-type parameters.*
  International E-Conference: A Selcuk Meeting, Selçuk University (Online, Octubre 2022).
- *New advances in greedy approximation theory.*
  Analysis Seminar - University of Illinois at Urbana-Champaign, EEUU (Abril 2023).
- *La condición de la avaricia en aproximación.*
  SUMA 2019, Mendoza (Argentina, Septiembre 2019).

## Organization of Sessions and Conferences

- **Parallel Session at the RSME Young Researchers Conference 2025**.
  *Numerical Methods for Applied Mathematics*, University of Bilbao, Spain (February 2025).
- **Parallel Session at the RSME Biennial Congress 2024**.
  *Numerical Methods and Computational Mathematics*, Pamplona, Spain (January 2024).
- **Applied Mathematics and Statistics Seminar, CEU San Pablo University**.
  *Member of the organizing committee (2020–2021).*

## Scientific Reviewing

Regular referee for the following journals and institutions: Mathematical Inequalities and Applications, Journal of Mathematical Analysis and Applications, Demonstratio Mathematica, Revista Matemática Complutense, Mathematics, Advances in Operator Theory, Zeitschrift für angewandte Mathematik und Physik, Annals of Functional Analysis, Jaen Journal on Approximation and Mathematical Review (AMS).

## Teaching

I have taught more than 2000 hours (spanish and english) at different universities and degree programs.

- **CUNEF University** (2022–2026): Mathematical Analysis II and III (Mathematical Engineering), Calculus II (Data Science), Statistical Inference (Mathematical Engineering), Data Analysis and Probability and Statistical Inference (Computer Engineering).
- **UTAMED** (2025–2026): Mathematics for AI and Deep Learning (Master in Artificial Intelligence).
- **CEU San Pablo University** (2020–2022): Mathematics I and II (Business Administration), Dynamical Systems and Optimization Theory (Economy).
- **European University** (2020–2022): Calculus II (Aerospace), Statistics (Physics), Biostatistics (Biomedical Engineering) and Mathematics I (Business Administration).

Additionally, I have supervised several undergraduate theses in the interaction between Mathematics and Economics, as well as a Master's thesis on greedy algorithms.